AI4101 · ARTIFICIAL INTELLIGENCE PRINCIPLES · LAB 1 REPORT

Teaching a Computer to See

Sample report — two fictional students

About this sample

This is a **sample** to show what a complete report looks like. The two students are fictional. It was prepared by the instructor with the help of an AI assistant, using a real Teachable Machine model: every number and screenshot below comes from that model.

The sample uses **photos from Wikimedia Commons**, because it was prepared without a camera. The photo credits are in `Lab1_Sample_Photos/credits.md`. **Your report must use real objects in front of your camera** (Part 2, condition 1). Where the photos changed an experiment, the sample says so.

The gold boxes are notes for you. They are not part of the report.

	Full name	Student ID
Partner 1	Student A (fictional)	000000001
Partner 2	Student B (fictional)	000000002

Lab section	1
Video link (Unlisted)	No video for this sample. Your report must include your link here.
Date submitted	26 September 2026

1. Our classifier

Our problem

Problem	Sorting writing tools
Default or our own?	Our own problem
Borderline object	A mechanical pencil

Our problem meets the four conditions: three classes of objects every student has (pen, pencil, eraser); at least two objects per class plus five kept aside for testing; a mechanical pencil as the borderline object; and no people. (In this sample, photos replace real objects — see the box at the top.)

1a. The idea in simple words

Our model looks at a picture of one writing tool and says whether it is a pen, a pencil or an eraser. We chose these three because every student has them, and because pens and pencils look alike, so the model has a real task. The model could be the "eye" of a machine that sorts lost stationery at a school desk.

1b. How we built it

Class name	Number of examples	Objects we used	What we varied
Pen	15	Ballpoint, gel and fountain pens	Colour, angle, background; pens with and without caps
Pencil	15	Wooden pencils of different colours	Colour, length, background
Eraser	16	White, pink and blue erasers	Shape, colour, background

Two problems we found later: one "pen" image is a drawing (an icon), not a photo, and one "pencil" image is a mechanical pencil. Section 6 explains why the second one matters.

> **Note for students**
> The table gives numbers, not words like "many". The report also admits two mistakes in the data. Hiding them would make the results in section 6 impossible to explain.

2. Testing on new objects

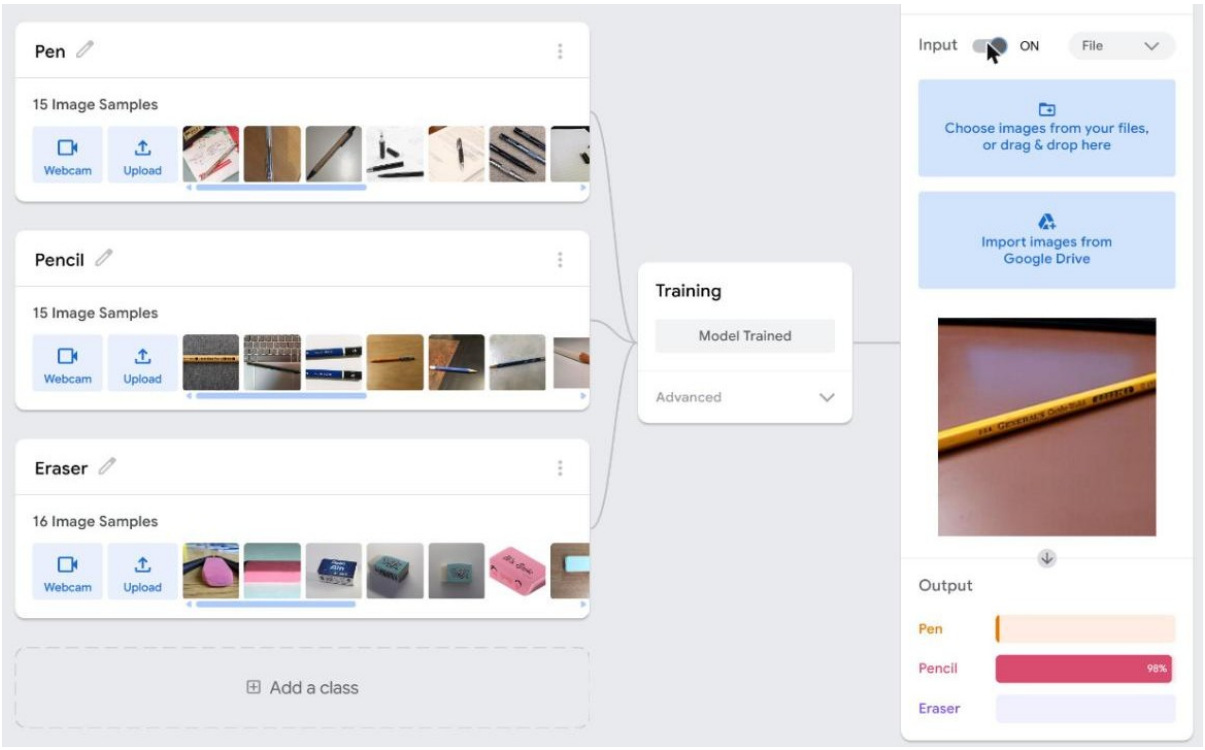
Objects that were **not** in our training data.

New object	True class	Model's answer	Confidence	Correct?
Black ballpoint pen on lined paper	Pen	Pencil	84%	No
Dip pen	Pen	Pen	92%	Yes
Yellow pencil on a brown table	Pencil	Pencil	98%	Yes
Wooden pencil on white	Pencil	Pencil	54%	Yes, but unsure
Small white eraser	Eraser	Eraser	100%	Yes

Result: 4 correct out of 5.

> **Note for students**
> Every test object is new. The report keeps the wrong answer and the unsure one (54%). A table where everything is correct usually means the test was too easy, or the objects were in the training data.

3. Screenshot 1 — a new object, classified correctly



A yellow pencil the model never saw in training. Output: Pencil 98%.

This is a yellow wooden pencil on a brown table. It is a fair test because this photo is not in our training data; the model has never seen this pencil, and it still chose Pencil with 98%.

4. Experiments

Experiment A — How much data?

Model	Examples per class	Correct out of 5 test objects
Part 2 model	15 to 16	4
Experiment A model	5	3

We used 5 examples per class instead of 10, because our Part 2 model had only 15. The Experiment A model was also less sure: it chose Pen with only 58% for the dip pen.

Experiment B — Change the background

Class	Part 2 test photo: answer (confidence)	Busy background: answer (confidence)
Pen	Pen (92%)	Pencil (81%) — wrong
Pencil	Pencil (98%)	Pencil (100%)
Eraser	Eraser (100%)	Eraser (100%)

New background we used:

Busy scenes: many coloured pens on coloured paper; a pencil on a patterned cloth; several erasers on a carpet. With photos we could not move the same object to a new background, so each busy scene is a different photo of the same class.

Experiment C — Something that belongs to no class

Object	Model's answer	Confidence
A silver key on white	Eraser	84%
A key on a wooden table	Pen	92%

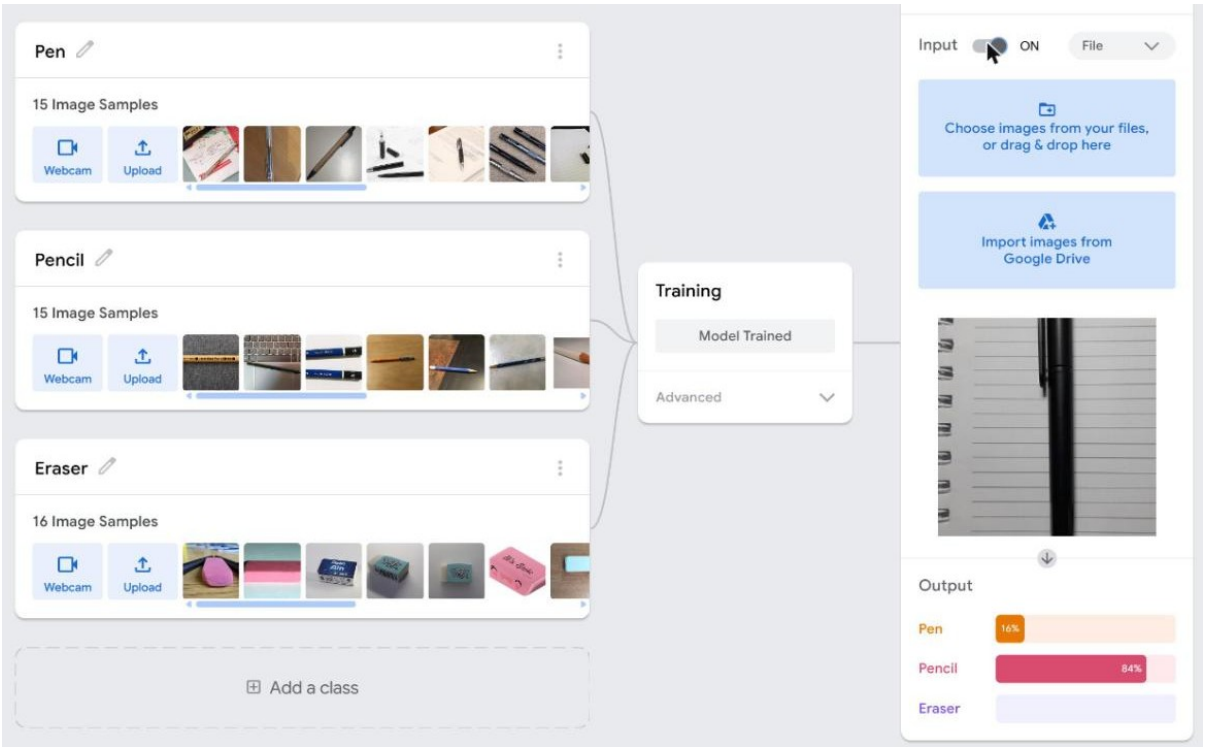
What the three experiments show

With fewer examples the model made more mistakes and was less sure. The busy picture of pens was called Pencil, so the model probably learned colours and backgrounds, not only shapes. The key is not a pen, a pencil or an eraser, but the model must choose one of our three classes: its percentages always add up to 100%, and it has no "none of these" class. So a high confidence (92%) does not mean a correct answer.

Note for students

Where the sample had to change an experiment (5 examples instead of 10; different photos instead of the same object), it says so and gives the reason. Do the same in your report.

5. Screenshot 2 — a failure

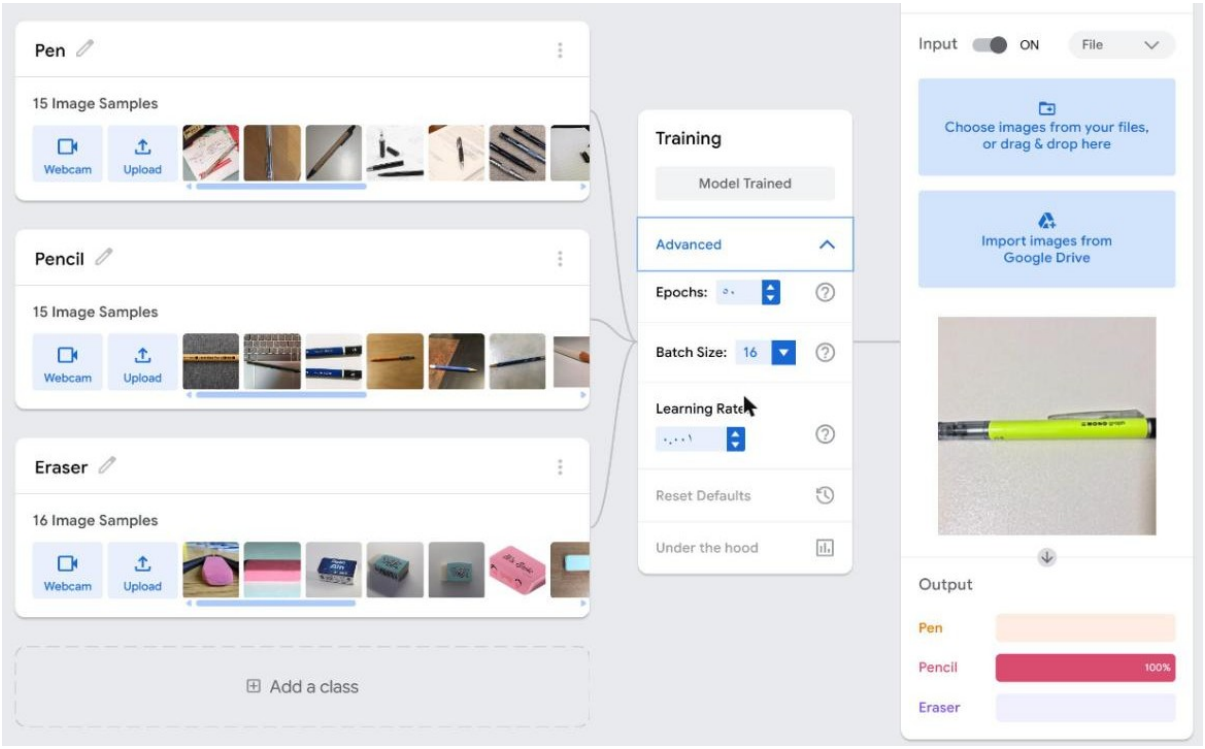


A black ballpoint pen on lined paper. Output: Pencil 84%.

This is from our Part 2 test. The model called a black ballpoint pen "Pencil" with 84%. Several of our pencil photos show a thin, dark object on light paper, while many of our pen photos show pens with colour and caps. The model probably learned "thin and dark on white paper" as a pencil pattern. More photos of plain black pens on paper would help.

6. The borderline case

Object	Model's answer	Confidence	Our correct label
Mechanical pencil (whole, yellow-green)	Pencil	100%	Pencil
Mechanical pencil (taken apart)	Pen	97%	Pencil



The whole mechanical pencil. Output: Pencil 100%.

Two questions

(a) At first we did not agree. Student A said Pen, because it has a click button and a plastic body like a pen. Student B said Pencil, because it writes with graphite and the writing can be erased. We decided on **Pencil**, because for a sorting machine the rule that matters is what the tool writes with. This is our decision as designers; the model cannot make it for us.

(b) When we checked our training data, we found that one of our 15 pencil photos is a mechanical pencil. So the model had already seen a similar object under the label Pencil, which explains 100% for the whole mechanical pencil. The second photo shows a mechanical pencil taken apart into pieces. The model never saw anything like that, and it chose Pen with 97%. The model's answer comes from what it saw, not from the rule we decided.

> **Note for students**
>
> The two questions are kept apart: (a) is the designers' decision about the label; (b) explains the model's answer from the training data. Checking the training data is what found the mechanical pencil hiding in the Pencil class.

7. Connect it to the course

Q1. Agent or not?

No. The Preview perceives an image (one percept) and shows percentages on the screen, but it does not act on its environment. In Module 1, an agent perceives its environment and acts upon that environment. The Preview is better seen as one part that a real agent could use, like the machine in Q2.

Q2. PEAS for our system

Our system:

A stationery sorting machine at a school's lost-and-found desk. A staff member puts one item under the camera, and the machine drops it into the correct box.

P — Performance measure	Each item ends up in the correct box; few items in the wrong box; items it is unsure about are given to a person instead of guessed.
E — Environment	The desk, the three boxes, the items people bring, the staff member, the light in the room.
A — Actuators	A small gate that drops the item into a box; a light that asks the staff member to check an item.
S — Sensors	A camera above the desk.

Note for students

P describes what counts as success (items in the right box), not how the machine works. "The camera recognises the item" would describe a method, not success.

Q3. Designer or data?

We decided the three classes and their names, which photos to use, the label of every photo, and that a mechanical pencil is a Pencil. The model learned the patterns in the photos that separate the three classes, such as colour, shape and background. It did not learn our rule about mechanical pencils; it only learned from the one example that happened to be in the data.

Q4. Who could it fail?

Coloured pens with many other things around them were called Pencil (Experiment B). At a school desk full of colourful items, the machine would put some pens in the pencil box. Students looking for a lost pen would look in the wrong box, and the staff would have to sort again. This comes from our data (Module 1, section 1.6): most of our pen photos were simple, so the model did not learn busy scenes.

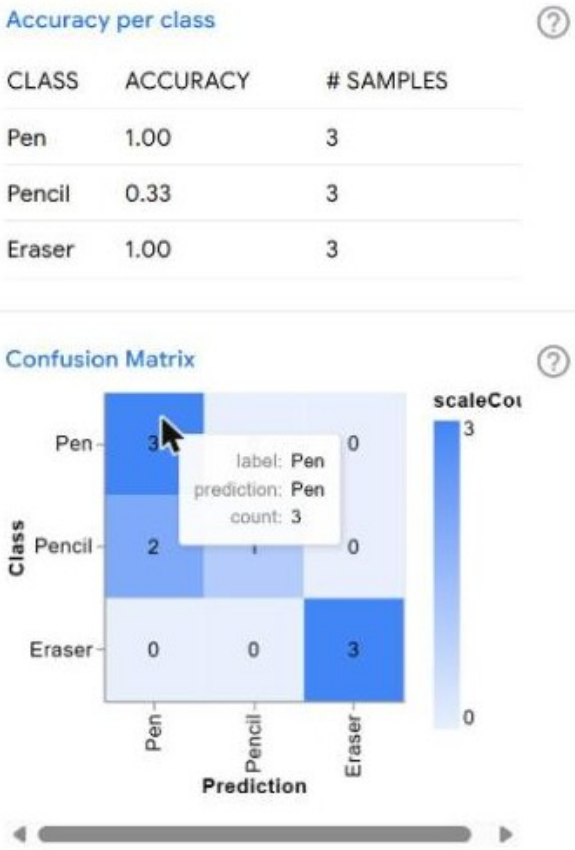
Q5. Be precise

This model recognises pens, pencils and erasers, one item at a time, from clear photos with a simple background; it is less reliable with busy backgrounds and with objects that are none of the three.

Note for students

The sentence says what the model recognises, from which input, and under which conditions — and it states the limits found in the experiments.

8. Optional — inside the model



Accuracy per class and confusion matrix (Advanced → Under the hood).

Our model confuses Pencil with Pen most: on the examples Teachable Machine kept aside, 2 of the 3 pencils were called Pen. Pens and pencils are both long and thin, and some of our pencil photos are dark, like pens.

9. Who did what, and AI tools used

Partner	What they did
Partner 1	Chose the pen and eraser photos, ran Experiments A and C, wrote sections 1 to 4.
Partner 2	Chose the pencil photos, ran Experiment B and the borderline case, wrote sections 5 to 7. We checked the whole report together.

We used an AI assistant to check the English of sections 1a and 6. The results, the screenshots and the ideas are our own.

> **Note for students**
> Section 9 is short and specific: who did which part, and what the AI tool was used for.